

Jashore University of Science and Technology
Faculty of Science and Engineering
Department of Computer Science and Engineering

Course Outline: Data Structures

Part A- Introduction

I. Course code: CSE 1201

II. Credit: 3

1.Course Summary

To become a successful computer professional, one needs to have in depth knowledge of data structures to apply them in problem solving and algorithms to analyze different solutions to problems.

2.Course Objectives

1. Apply different data structures to problem solving
2. Explain different algorithms of solutions to problems.
3. Analyze different problem solutions

3.Course Learning Outcomes

1. Implements basic data structures such as Stacks, Queues, Graph and Trees.
2. Uses Big 'O' notation to express algorithmic running time.
3. Describes and analyzes elementary sorting algorithms such as Selection sort, Bubble sort, Insertion sort, and Shell sort.
4. Understands and restates the fundamentals of basic data structures.
5. Designs and analyzes simple algorithms.

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
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Week 1 Week 2 Week 3	Stacks, Queues and Recursion: Fundamentals, Different types of stacks and queues: Circular, dequeues etc.; evaluation of expressions, multiple stacks and queues; Recursion: Direct and indirect recursion, depth of recursion; Simulation of Recursion: Removal of Recursion; Towers of Hanoi.	Apply stacks, queues and recursion; Differentiate between stacks and queues; Evaluate expressions using stacks; Analyze recursive functions;	CLO1	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 4 Week 5 Week 6	Elements of Graphs and Trees: Graph Terminology, Paths and Circuits, Connectedness, Matrix Representation of Graph and Isomorphism of graphs. Trees, Rooted trees, Path Lengths in Rooted Trees.	Construct graphs and trees; Differentiate between graphs and trees	CLO1	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 7 Week 8	Linked Lists: Single linked lists, Linked stacks and queues, the storage pool, polynomial addition, equivalence relations, sparse matrices, doubly linked lists and dynamic storage management, generalized lists, garbage collection and compaction.	Construct linked lists; Apply linked lists to stacks and queues; Analyze memory allocation;	CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 9 Week 10	Extended binary trees: 2-trees, internal and external path lengths, Huffman codes/algorithm; Threaded binary trees, binary tree representation of trees; Application of Trees: Set representation, decision trees, game trees; Counting binary trees.	Construct and apply extended binary trees	CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

Week 11 Week 12	Sorting: Searching, bubble sort, shell sort, insertion sort, selection sort, quick sort, heap sort, 2-way merge sort, sorting on several keys, practical considerations of internal sorting. searching, hash techniques.	Apply sorting algorithms; Distinguish among sorting algorithms;	CLO3	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 13 Week 14	Algorithms: Techniques for analysis of algorithms; Algorithmic Techniques: divide-and-conquer, greedy method, dynamic programming, backtracking, branch and bound; Flow algorithms. Topological sorting; Connected components; spanning trees; Shortest paths; Algebraic simplification and transformations; Lower bound theory; NP-completeness; NP-hard and NP-complete problems; Approximation Algorithms; Introduction to parallel algorithms.	Apply algorithmic techniques; Analyze algorithms;	CLO2, CLO5	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	Description
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Assignment	Type of home work assessment, students are given specific tasks and instructed to complete them within a given period of time.
Practical	Mostly related to laboratory works or by implementing in real code in either exercise book or in programming IDE

Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions
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Mapping of CLOs to Assessment

CIE- Continuous Internal Evaluation (28 Marks):

Bloom's Category Marks (out of 20)	Test (10)	Assignm ent (5)	Quizzes (5)	External Participation in Curricular/Co-Curricular Activities
Remember	2			
Understand	5			
Apply	3			
Analyze	2			
Evaluate	4			
Create	4			

SEE-Semester End Examination (72 Marks)

Bloom's Category	Test
Remember	05
Understand	15
Apply	20
Analyze	10
Evaluate	10
Create	12

Mapping of CLOs to PLOs

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9
CLO1	1	0	0	0	0	1	0	0	0
CLO2	0	1	0	0	0	0	1	0	0

CLO3	0	0	1	0	0	0	0	1	1
CLO4	0	0	0	1	0	0	0	1	0
CLO5	0	0	0	0	1	0	0	0	1

Part D- Resources

Textbooks

1. Theory and Problem of Data Structures by S. Lipschutz
2. Data Structure by E. Horowitz

Reference Books

1. The Art of Computer Programming, Vol. 1, Fundamental Algorithms by D. E. Knuth
2. The Art of Computer Programming, Vol. 2, Fundamental Algorithms by D. E. Knuth
3. The Art of Computer Programming, Vol. 3, Fundamental Algorithms by D. E. Knuth

Jashore University of Science and Technology
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Department of Computer Science and Engineering

Course Outline: Data Structures Lab

Part A- Introduction

I. Course code: CSE 1202

II. Credit: 1.5

1. Course Summary

To become a successful computer professional, one needs to have in depth knowledge of data structures to apply them in problem solving and algorithms to analyze different solutions to problems.

2. Course Objectives

1. Apply different data structures to problem solving
2. Explain different algorithms of solutions to problems.
3. Analyze different problem solutions

3. Course Learning Outcomes

- CLO1.** Implements basic data structures such as Stacks, Queues, Graph and Trees.
- CLO2.** Uses Big 'O' notation to express algorithmic running time.
- CLO3.** Describes and analyzes elementary sorting algorithms such as Selection sort, Bubble sort, Insertion sort, and Shell sort.
- CLO4.** Understands and restates the fundamentals of basic data structures.
- CLO5.** Designs and analyzes simple algorithms.

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1 Week 2 Week 3	Stacks, Queues and Recursion: Fundamentals, Different types of stacks and queues: Circular, dequeues etc.; evaluation of expressions, multiple stacks and queues; Recursion: Direct and indirect recursion, depth of recursion; Simulation of Recursion: Removal of Recursion; Towers of Hanoi.	Apply stacks, queues and recursion; Differentiate between stacks and queues; Evaluate expressions using stacks; Analyze recursive functions;	CLO1	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 4 Week 5 Week 6	Elements of Graphs and Trees: Graph Terminology, Paths and Circuits, Connectedness, Matrix Representation of Graph and Isomorphism of graphs. Trees, Rooted trees, Path Lengths in Rooted Trees.	Construct graphs and trees; Differentiate between graphs and trees	CLO1	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 7 Week 8	Linked Lists: Single linked lists, Linked stacks and queues, the storage pool, polynomial addition, equivalence relations, sparse matrices, doubly linked	Construct linked lists; Apply linked lists to stacks and queues; Analyze memory allocation;	CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

	lists and dynamic storage management, generalized lists, garbage collection and compaction.				
Week 9 Week 10	Extended binary trees: 2-trees, internal and external path lengths, Huffman codes/algorithm; Threaded binary trees, binary tree representation of trees; Application of Trees: Set representation, decision trees, game trees; Counting binary trees.	Construct and apply extended binary trees	CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 11 Week 12	Sorting: Searching, bubble sort, shell sort, insertion sort, selection sort, quick sort, heap sort, 2-way merge sort, sorting on several keys, practical considerations of internal sorting, searching, hash techniques.	Apply sorting algorithms; Distinguish among sorting algorithms;	CLO3	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 13 Week 14	Algorithms: Techniques for analysis of algorithms; Algorithmic Techniques: divide-and-conquer, greedy method, dynamic programming, backtracking, branch and bound; Flow algorithms. Topological sorting;	Apply algorithmic techniques; Analyze algorithms;	CLO2, CLO5	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

	Connected components; spanning trees; Shortest paths; Algebraic simplification and transformations; Lower bound theory; NP-completeness; NP-hard and NP-complete problems; Approximation Algorithms; Introduction to parallel algorithms.				
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Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	Description
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Assignment	Type of home work assessment, students are given specific tasks and instructed to complete them within a given period of time.
Practical	Mostly related to laboratory works or by implementing in real code in either exercise book or in programming IDE
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

CIE- Continuous Internal Evaluation (40 Marks):

Bloom's Category Marks (out of 20)	Test (30)	Attendance (10)	External Participation in Curricular/Co- Curricular Activities
Remember	2		
Understand	5		
Apply	13		
Analyze	2		
Evaluate	4		
Create	4		

SE- Lab Final Exam (60 marks)

Bloom's Category	Test
Remember	5
Understand	10
Apply	25
Analyze	10
Evaluate	5
Create	5

Mapping of CLOs to PLOs

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9
CLO1	1	0	0	0	0	1	0	0	0
CLO2	0	1	0	0	0	0	1	0	0
CLO3	0	0	1	0	0	0	0	1	1
CLO4	0	0	0	1	0	0	0	1	0
CLO5	0	0	0	0	1	0	0	0	1

Part D- Resources

Textbooks

1. Theory and Problem of Data Structures by S. Lipschutz
2. Data Structure by E. Horowitz

Reference Books

1. The Art of Computer Programming, Vol. 1, Fundamental Algorithms by D. E. Knuth
2. The Art of Computer Programming, Vol. 2, Fundamental Algorithms by D. E. Knuth
3. The Art of Computer Programming, Vol. 3, Fundamental Algorithms by D. E. Knuth

Jashore University of Science and Technology
Faculty of Science and Engineering
Department of Computer Science and Engineering

Course Outline: Digital Logic Design

Part A- Introduction

I. Course code: CSE 1203

II. Credit: 3

1. Course Summary

A computer engineer needs to know about number system, logic design and the basic building blocks used in digital systems, in particular digital computers.

2. Course Objectives

1. To learn about basic concept on number system, universal gates and truth tables.
2. To know about Boolean function and De-Morgan, canonical forms and minimization techniques
3. To learn about combinational and sequential circuits and basic flip flops
4. To know about synchronous and asynchronous counters

3. Course Learning Outcomes

- CLO1.** Identify and explain fundamental concepts of digital logic design including basic and universal gates, number systems, binary coded systems, basic components of combinational and sequential circuits
- CLO2.** Demonstrate the acquired knowledge to apply techniques related to the design and analysis of digital electronic circuits including Boolean algebra and multi-variable Karnaugh map methods

- CLO3.** Analyze small-scale combinational and sequential digital circuits
- CLO4.** Design small-scale combinational and synchronous sequential digital circuit using Boolean algebra and K-maps

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1 Week 2 Week 3	Number systems and codes. Digital logic: Boolean algebra, De-Morgan's law, Logic gates and their truth tables.	Explain number systems Apply De-Morgan law for the Boolean function State and explain De-Morgan	CLO1, CLO2	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 4 Week 5 Week 6	Canonical forms, combinational logic circuits, minimization techniques.	Evaluate canonical forms Apply minimization techniques Explain combinational logic circuits	CLO1, CLO2	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 7 Week 8	Arithmetic and data handling logic circuit, decoders and encoders, Multiplexers and De-multiplexers.	Determine decoder and encoder function Differentiate MUX and DEMUX Draw data handling logic circuit	CLO3,	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 9 Week 10	Combinational Circuit design, Flip-flops, race around	Explain flip flops and race problems	CLO3, CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

	problems.				
Week 11 Week 12	Counters: Asynchronous and Synchronous counters and their applications. Synchronous and asynchronous logic design: state diagram, Mealy and Moore machine.	Design synchronous and asynchronous circuits Apply state diagram for logic circuits, State and explain Mealy and Moore machine	CLO3, CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 13 Week 14	State minimization and assignments. Pulse mode logic, Fundamental mode logic design.	Explain state minimization and assignments Demonstrate pulse mode logic and mode logic design	CLO3, CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	Description
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Assignment	Type of home work assessment, students are given specific tasks and instructed to complete them within a given period of time.
Practical	Mostly related to laboratory works or by implementing in real code in either exercise book or in programming IDE
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

CIE- Continuous Internal Evaluation (28 Marks):

Bloom's Category Marks (out of 20)	Test (10)	Assignment (5)	Quizzes (5)	External Participation in Curricular/Co-Curricular Activities
Remember	2			
Understand	5			
Apply	3			
Analyze	2			
Evaluate	4			
Create	4			

SEE-Semester End Examination (72 Marks)

Bloom's Category	Test
Remember	05
Understand	15
Apply	20
Analyze	10
Evaluate	10
Create	12

Mapping of CLOs to PLOs

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9
CLO1	1	0	0	0	0	1	0	0	0
CLO2	0	1	0	0	0	0	1	0	0
CLO3	0	0	1	0	0	0	0	1	1
CLO4	0	0	0	1	0	0	0	1	0

Part D- Resources

Textbooks

1. Digital logic and Computer Design by M. Morris Mano
2. Digital System Analysis by Tocci

Reference Books

1. Principles of Electronic by D. E. Knuth
2. Electronic Devices and Circuit Theory by R.L. Boylestad

Jashore University of Science and Technology
Faculty of Science and Engineering
Department of Computer Science and Engineering

Course Outline: Digital Logic Design Lab

Part A- Introduction

III. Course code: CSE 1204

IV. Credit: 1.5

4. Course Summary

A computer engineer needs to know about number system, logic design and the basic building blocks used in digital systems, in particular digital computers.

5. Course Objectives

1. To learn about basic concept on number system, universal gates and truth tables.
2. To know about Boolean function and De-Morgan, canonical forms and minimization techniques
3. To learn about combinational and sequential circuits and basic flip flops
4. To know about synchronous and asynchronous counters

6. Course Learning Outcomes

- CLO1.** Apply and explain fundamental concepts of digital logic design including basic and universal gates, number systems, binary coded systems, basic components of combinational and sequential circuits
- CLO2.** Implement the acquired knowledge to apply techniques related to the design and analysis of digital electronic circuits including Boolean algebra and multi-variable Karnaugh map methods

- CLO3.** Analyze small-scale combinational and sequential digital circuits
- CLO4.** Design small-scale combinational and synchronous sequential digital circuit using Boolean algebra and K-maps

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1 Week 2 Week 3	Number systems and codes. Digital logic: Boolean algebra, De-Morgan's law, Logic gates and their truth tables.	Explain number systems Apply De-Morgan law for the Boolean function State and explain De-Morgan	CLO1, CLO2	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 4 Week 5 Week 6	Canonical forms, combinational logic circuits, minimization techniques.	Evaluate canonical forms Apply minimization techniques Explain combinational logic circuits	CLO1, CLO2	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 7 Week 8	Arithmetic and data handling logic circuit, decoders and encoders, Multiplexers and De-multiplexers.	Determine decoder and encoder function Differentiate MUX and DEMUX Draw data handling logic circuit	CLO3,	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

Week 9 Week 10	Combinational Circuit design, Flip-flops, race around problems.	Explain flip flops and race problems	CLO3, CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 11 Week 12	Counters: Asynchronous and Synchronous counters and their applications. Synchronous and asynchronous logic design: state diagram, Mealy and Moore machine.	Design synchronous and asynchronous circuits Apply state diagram for logic circuits, State and explain Mealy and Moore machine	CLO3, CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 13 Week 14	State minimization and assignments. Pulse mode logic, Fundamental mode logic design.	Explain state minimization and assignments Demonstrate pulse mode logic and mode logic design	CLO3, CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	Description
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Assignment	Type of home work assessment, students are given specific tasks and instructed to complete them within a given period of time.
Practical	Mostly related to laboratory works or by implementing in real code in either exercise book or in programming IDE
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

CIE- Continuous Internal Evaluation (40 Marks):

Bloom's Category Marks (out of 20)	Test (30)	Attendance (10)	External Participation in Curricular/Co-Curricular Activities
Remember	2		
Understand	5		
Apply	13		
Analyze	2		
Evaluate	4		
Create	4		

SE- Lab Final Exam (60 marks)

Bloom's Category	Test
Remember	5
Understand	10
Apply	25

Analyze	10
Evaluate	5
Create	5

Mapping of CLOs to PLOs

CLOs	PLO10	PLO11	PLO12	PLO13	PLO14	PLO15	PLO16	PLO17	PLO18
CLO1	1	0	0	0	0	1	0	0	0
CLO2	0	1	0	0	0	0	1	0	0
CLO3	0	0	1	0	0	0	0	1	1
CLO4	0	0	0	1	0	0	0	1	0

Part D- Resources

Textbooks

3. Digital logic and Computer Design by M. Morris Mano
4. Digital System Analysis by Tocci

Reference Books

3. Principles of Electronic by D. E. Knuth
4. Electronic Devices and Circuit Theory by R.L. Boylestad

Jashore University of Science and Technology
Faculty of Science and Engineering
Department of Computer Science and Engineering

Course Outline: Engineering Drawing

Part A- Introduction

V. Course code: CSE 1206

VI. Credit: 1

7. Course Summary

Capability of computer aided design and drawing is essential for an engineer in order to be able to design objects using some drawing software..

8. Course Objectives

1. Understand scaling of drawing different objects.
2. Recognize objects from different views.
3. Design structures using drawing software

9. Course Learning Outcomes

- CLO5. Ability** to read basic engineering drawing.
- CLO6. Ability** to draw basic drawing objects on standard drawing sheets.
- CLO7. Apply** engineering drawing skills using Auto CAD tool.
- CLO8. Respond** as instructed while working in groups.

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1- Week 6	Familiarization with different logic gates and flip-flops.	Apply scales Identify views	CLO1 CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 7- Week 10	Missing line, Auxiliary view, Detail and assembly drawing.	Find out missing lines. Apply auxiliary view and assembly drawing	CLO2 CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 11- Week 14	Flip-flops, Sequential circuit, Counter, Combinational Circuit.	Design objects using CAD.	CLO3, CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	Description
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Assignment	Type of home work assessment, students are given specific tasks and instructed to complete them within a given period of time.
Practical	Mostly related to laboratory works or by implementing in real code in either exercise book or in programming IDE
Exam	Includes set of programming problems to solve within a timeline

CIE- Continuous Internal Evaluation (40 Marks):

Bloom's Category Marks (out of 20)	Test (30)	Attendance (10)	External Participation in Curricular/Co- Curricular Activities
Remember	2		
Understand	5		
Apply	13		
Analyze	2		
Evaluate	4		
Create	4		

SE- Lab Final Exam (60 marks)

Bloom's Category	Test
Remember	5
Understand	10
Apply	25
Analyze	10
Evaluate	5
Create	5

Mapping of CLOs to PLOs

CLOs	PLO19	PLO20	PLO21	PLO22	PLO23	PLO24	PLO25	PLO26	PLO27
CLO1	1	0	0	0	0	1	0	0	0
CLO2	0	1	0	0	0	0	1	0	0
CLO3	0	0	1	0	0	0	0	1	1
CLO4	0	0	0	1	0	0	0	1	0

Part D- Resources

Textbooks

5. Mastering AutoCAD 2008 and AutoCAD LT 2008 by Wiley
6. Computer Aided Design and manufacturing by M. Groover and E. Zimmers

Reference Books

5. CAD/CAM Robotics and Factories of the Future by S. Narayanan
6. Computer Aided Manufacturing by TienChien Chang

Jashore University of Science and Technology
Faculty of Science and Engineering
Department of Computer Science and Engineering
Course Outline: Electronic Devices and Circuits

Part A- Introduction

I. Course code: EEE 1201

II. Credit: 3

1. Course Summary

To learn basic concepts about semiconductor physics.

To know about formation of semiconductor diodes, transistors, field effect transistors, Oscillators and their working principle.

To know about power electronics devices and its operation.

A computer engineer needs to know about number system, logic design and the basic building blocks used in digital systems, in particular digital computers.

2. Course Objectives

1. To learn basic concepts about semiconductor physics.
2. To know about formation of semiconductor diodes, transistors, field effect transistors, Oscillators and their working principle.
3. To know about power electronics devices and its operation.

3. Course Learning Outcomes

- CLO1.** learn basic concepts about semiconductor physics
- CLO2.** Know about formation of semiconductor diodes, transistors
- CLO3.** know about power electronics devices and its operation
- CLO4.** Apply amplifier in different circuits

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1 Week 2 Week 3	Theory of Semiconductors: Electronic structure of the elements, crystalline and amorphous solids, different types of crystal, band theory of solids, structure of silicon and germanium.	Define and discuss semiconductor physics and semiconductors.	CLO1, CLO2	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 4 Week 5 Week 6	Intrinsic and Extrinsic Semiconductors: N and P type semiconductor, carrier densities, generation and recombination of excess carriers, carrier life time, continuity equation.	Define and discuss semiconductor physics and semiconductors.	CLO1, CLO2	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 7 Week 8	Diode circuits: The PN junction, biasing and V-I characteristics of diodes, rectifier concept, half wave and full wave rectifiers, Zener diode and voltage	Define diodes	CLO3,	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

	regulators.				
Week 9 Week 10	Bipolar transistor: Junction transistors, principles of operation, biasing, characteristics in different configurations (CE, CB & CC), DC and AC load line, transistor equivalent circuits. Gain and impedance, Analysis of small signal low frequency transistor amplifier by using h-parameters.	Draw and describe diode clipping and clamping circuits.	CLO3, CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 11 Week 12	Field effect transistor: Construction of JFET, its parameters, biasing, characteristics and principles of operation, different types of MOSFET and its operation and characteristics. Amplifier: Voltage and current amplifiers of different configurations, RC coupled amplifier, operational amplifiers (OPAMPS), linear applications of OPAMPS, gain, input and output	Define and explain Operational Amplifiers and FET	CLO3, CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

	impedance.				
Week 13 Week 14	Capacitor and Capacitance: Capacitance, Parallel plate capacitor, cylindrical capacitor, spherical capacitor, capacitors in series & parallel, energy stored in a capacitor, transformers.	Classify Oscillators.	CLO3, CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	Description
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
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Understand	5			
Apply	3			
Analyze	2			
Evaluate	4			
Create	4			

SEE-Semester End Examination (72 Marks)

Bloom's Category	Test
Remember	05
Understand	15
Apply	20
Analyze	10
Evaluate	10
Create	12

Mapping of CLOs to PLOs

CLOs	PLO28	PLO29	PLO30	PLO31	PLO32	PLO33	PLO34	PLO35	PLO36
CLO1	1	0	0	0	0	1	0	0	0
CLO2	0	1	0	0	0	0	1	0	0
CLO3	0	0	1	0	0	0	0	1	1
CLO4	0	0	0	1	0	0	0	1	0

Part D- Resources

Textbooks

1. Basic Electronics (Solid State).
2. Electronic Devices and Circuit Theory.

Reference Books

1. Hand Book of Electronics.
2. Electronic Devices and Circuits.

Jashore University of Science and Technology
Faculty of Science and Engineering
Department of Computer Science and Engineering
Course Outline: Electronic Devices and Circuits Lab

Part A- Introduction

I. Course code: EEE 1202

II. Credit: 1.5

1. Course Summary

To learn basic concepts about semiconductor physics.

To know about formation of semiconductor diodes, transistors, field effect transistors, Oscillators and their working principle.

To know about power electronics devices and its operation.

A computer engineer needs to know about number system, logic design and the basic building blocks used in digital systems, in particular digital computers.

2. Course Objectives

1. To learn basic concepts about semiconductor physics.
2. To know about formation of semiconductor diodes, transistors, field effect transistors, Oscillators and their working principle.
3. To know about power electronics devices and its operation.

3. Course Learning Outcomes

- CLO1.** Implement basic concepts about semiconductor physics
- CLO2.** Demonstrate about formation of semiconductor diodes, transistors
- CLO3.** Apply about power electronics devices and its operation
- CLO4.** Apply amplifier in different circuits

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1 Week 2 Week 3	Theory of Semiconductors: Electronic structure of the elements, crystalline and amorphous solids, different types of crystal, band theory of solids, structure of silicon and germanium.	Define and discuss semiconductor physics and semiconductors.	CLO1, CLO2	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 4 Week 5 Week 6	Intrinsic and Extrinsic Semiconductors: N and P type semiconductor, carrier densities, generation and recombination of excess carriers, carrier life time, continuity equation.	Define and discuss semiconductor physics and semiconductors.	CLO1, CLO2	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 7 Week 8	Diode circuits: The PN junction, biasing and V-I characteristics of diodes, rectifier concept, half wave and full wave rectifiers, Zener diode and voltage regulators.	Define diodes	CLO3,	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 9 Week 10	Bipolar transistor: Junction transistors, principles of operation, biasing, characteristics in different configurations (CE, CB & CC), DC and AC load line, transistor equivalent circuits. Gain and impedance, Analysis of small signal low frequency transistor amplifier by using h -parameters.	Draw and describe diode clipping and clamping circuits.	CLO3, CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

Week 11 Week 12	Field effect transistor: Construction of JFET, its parameters, biasing, characteristics and principles of operation, different types of MOSFET and its operation and characteristics. Amplifier: Voltage and current amplifiers of different configurations, RC coupled amplifier, operational amplifiers (OPAMPS), linear applications of OPAMPS, gain, input and output impedance.	Define and explain Operational Amplifiers and FET	CLO3, CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 13 Week 14	Capacitor and Capacitance: Capacitance, Parallel plate capacitor, cylindrical capacitor, spherical capacitor, capacitors in series & parallel, energy stored in a capacitor, transformers.	Classify Oscillators.	CLO3, CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	Description
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Assignment	Type of home work assessment, students are given specific tasks and instructed to complete them within a given period of time.
Practical	Mostly related to laboratory works or by implementing in real code in either exercise book or in programming IDE
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

CIE- Continuous Internal Evaluation (40 Marks):

Bloom's Category Marks (out of 20)	Test (30)	Attendance (10)	External Participation in Curricular/Co- Curricular Activities
Remember	2		
Understand	5		
Apply	13		
Analyze	2		
Evaluate	4		
Create	4		

SE- Lab Final Exam (60 marks)

Bloom's Category	Test
Remember	5
Understand	10
Apply	25
Analyze	10
Evaluate	5
Create	5

Mapping of CLOs to PLOs

CLOs	PLO37	PLO38	PLO39	PLO40	PLO41	PLO42	PLO43	PLO44	PLO45
CLO1	1	0	0	0	0	1	0	0	0
CLO2	0	1	0	0	0	0	1	0	0
CLO3	0	0	1	0	0	0	0	1	1
CLO4	0	0	0	1	0	0	0	1	0

Part D- Resources

Textbooks

1. Basic Electronics (Solid State).

2. Electronic Devices and Circuit Theory.

Reference Books

1. Hand Book of Electronics.
2. Electronic Devices and Circuits.

Jashore University of Science and Technology
Faculty of Science and Engineering
Department of Computer Science and Engineering

Differential Equations and Geometry

Course Code: MATH 1201
Exam Hours : 03

Credit: 3.0
SEE Marks : 72
CIE Marks : 28

Course Learning Outcomes: *After the successful completion of the course, students will be able to:*

CLO1	Understand and explain formation and classification of differential equation.
CLO2	Explain basic concepts of first order and first degree differential equation.
CLO3	Understand and explain basic concepts higher order differential equation and their solution process.
CLO4	Learn and explain coordinates system, transformations of coordinates.
CLO5	Understand and explain straight lines, pair of straight lines.
CLO6	Explain and expand General Equation of Second Degree.
CLO7	Learn and determine Circles, System of circles, Tangent, Normal.

S N	Course Content	Hrs	CLOs
1.	Definitions and classifications of differential equations, problems and solutions, Formation of differential equation, Existence and uniqueness theorem (Statement and application only).	6	CLO1
2.	Separable and homogeneous equations, Exact equation, Integrating factor, Equations made exact by integrating factor, First order linear equation, Bernoulli equation, Riccati equation.	6	CLO2
3.	First order higher degree equations-solvable for x, y and p, Clairaut's equation, Singular solutions, Orthogonal and oblique trajectories. Higher order linear homogeneous equation with constant coefficients, Reduction of order, Basic theorems, Linear non-homogeneous equation with constant coefficients, Method of undetermined coefficients.	6	CLO3
4.	Co-ordinates: Coordinates system, Transformations of coordinates	5	CLO4

5.	Transformations: Change of axes, Translation, Invariants, Analyzing an equation using a rotation (identify and sketch).	6	CLO5
6.	Straight Lines: Straight lines, Pair of straight lines.	3	CLO5
7.	General Equation of Second Degree: Reduction to standard form and Identification of conics.	4	CLO6
8.	Circles: Circles, System of circles, Tangent, Normal.	3	CLO7

Recommended References:

Anton, H., Bivens, I. and Davis, S.	: Calculus
Das and Mukherjee	: Differential Calculus
Thomson & Finne.	: Calculus
Swokowski. E. W	: Calculus
Shanti Narayan	: Differential Calculus

ASSESSMENT PATTERN

CIE- Continuous Internal Evaluation (28Marks)

Bloom's Category Marks (out of 20)	Tests (10)	Assignments (05)	Quizzes (05)	External Participation in Curricular/Co-Curricular Activities(08)
Remember			05	
Understand		03		
Apply	04			8
Analyze	03			
Evaluate	03			
Create		02		

SEE- Semester End Examination (72 Marks)

Bloom's Category	Test
Remember	15
Understand	15
Apply	15
Analyze	15
Evaluate	10
Create	02

Jashore University of Science and Technology
Faculty of Science and Engineering
Department of Computer Science and Engineering

Course Outline: Physics for Engineer

Part A- Introduction

I. Course code: PHY 1201

II. Credit: 2

1. Course Summary

Someone wants to develop his/her career as a computer engineer needs to know the basic theories and principles of physics to generate problem solving, analytical, mathematical and solution finding skills; this course will equip him/her with the concepts of Kinetic Theory of Gases, Heat and Thermodynamics, Simple Harmonic Motion, Wave Motion, Sound Waves, Acoustics and Electrostatics.

2. Course Objectives

1. Apply the concepts, ideas and methods of Physics required to solve problems in engineering studies.
2. Acquire knowledge about different laws and models of Physics, which will develop analytical capabilities among them.
3. Apply the laws and skills in higher studies or research areas.
4. Understand the origins in thermodynamics, electronics and Acoustics.
5. Explain everyday things happening around us.

3. Course Learning Outcomes

- CLO1.** Apply the concepts, ideas and methods of Physics
- CLO2.** Apply the laws and skills in higher studies
- CLO3.** Acquire knowledge about different laws and models of Physics
- CLO4.** Explain everyday things happening around us

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1 Week 2 Week 3	Kinetic theory of gases: Deduction of gas law, Principle of equipartition of energy, Equation of state – Andrew’s Experiment, Vander Waals equation, Critical Constants, Transmission of heat – conduction, convection and radiation.	Define critical constants	CLO1, CLO2	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 4 Week 5 Week 6	Heat and Thermodynamics: 1st law of Thermodynamics, Internal energy, Specific heats of gases, Work done by expanding gas, Elasticity of a perfect gas, 2nd law of thermodynamics, Carnot’s cycle, Efficiency of heat engines, Absolute scale of temperature, Entropy and its physical concepts, Maxwell’s thermodynamic relations, Statistical Mechanics	Explain work done by expanding gas	CLO1, CLO2	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 7 Week 8	Simple Harmonic Motion: Simple harmonic motion, Combination of S.H.M. and Lissajous figures, Damped Oscillations, Forced oscillations, Resonance, Vibrations of membranes and columns. Electrostatics Charge and matter, Coulomb’s Law, The electric field, Gauss’s Law, Electrical potential, Capacitance and Resistance, Ohmic and Non ohmic material,	Draw the mismatch between Damped Oscillation and Forced Oscillation	CLO3,	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

	Variation of resistance with temperature.				
Week 9 Week 10	Wave motion: Travelling waves, Principle of superposition, wave velocity, Group velocity, Phase velocity, Power and intensity in wave motion, Interference, diffraction and transmission of waves, Standing Waves.	List the types of velocity play role in wave motion and interpret each with respective example	CLO3, CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 11 Week 12	Sound waves: Audible, Ultrasonic, Infrasonic and Supersonic waves, Propagation and speed of longitudinal waves, Travelling longitudinal waves, Standing longitudinal waves, Vibrating systems and sources of sound, Beats, The Doppler's effect	Differentiate between Ohmic and Non Ohmic material	CLO3, CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 13 Week 14	Acoustics: Re-vibration, Noise insulation and reduction, Compound absorption, sound distribution, Room acoustics, Recording.	Explain state minimization and assignments Demonstrate pulse mode logic and mode logic design	CLO3, CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	Description
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Assignment	Type of home work assessment, students are given specific tasks and instructed to complete them within a given period of time.
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Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

CIE- Continuous Internal Evaluation (28 Marks):

Bloom's Category Marks (out of 20)	Test (10)	Assignment (5)	Quizzes (5)	External Participation in Curricular/Co- Curricular Activities
Remember	2			
Understand	5			
Apply	3			
Analyze	2			
Evaluate	4			
Create	4			

SEE-Semester End Examination (72 Marks)

Bloom's Category	Test
Remember	05
Understand	15
Apply	20
Analyze	10
Evaluate	10
Create	12

Mapping of CLOs to PLOs

CLOs	PLO46	PLO47	PLO48	PLO49	PLO50	PLO51	PLO52	PLO53	PLO54
CLO1	1	0	0	0	0	1	0	0	0
CLO2	0	1	0	0	0	0	1	0	0
CLO3	0	0	1	0	0	0	0	1	1
CLO4	0	0	0	1	0	0	0	1	0

Part D- Resources

Textbooks

1. Equilibrium Thermodynamics by Adkins
2. Waves and Oscillations by Brijlal
3. Fundamentals of physics by Halliday and Rensick